Vivekanand Education Society's
Institute of Technology
(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)
NAAC accredited with 'A' grade

PROJECT REPORT ON

BioToken:

A Decentralized Framework for Reagent Provenance and Verification using Zero-Knowledge Proofs and Blockchain

SUBMITTED IN FULFILLMENT OF THE REQUIREMENT FOR SEMESTER VI OF

T.E.(Information Technology)

SUBMITTED BY

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Institute of Technology

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This is to certify that project entitled

"Biotoken: A Decentralized Framework for Reagent provenance and Verification using Zero-knowledge proof and Blockchain"

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Abstract

BioToken is a decentralized framework designed to address critical challenges in the global biochemical reagent supply chain, including the inability to verify manufacturer authenticity without exposing proprietary formulas, the widespread issue of empty-bottle fraud, and the risk of undetected quality degradation due to improper storage conditions. Unlike traditional solutions that rely on centralized databases or manual auditing, BioToken leverages Polygon PoS blockchain and ERC-721 NFTs to convert high-value reagents into secure, traceable digital assets with an immutable chain of custody. The system integrates Zero-Knowledge Proofs (ZK-SNARKs) to enable privacy-preserving verification, ensuring sensitive information remains protected while authenticity is guaranteed. An AI-based anomaly detection model, trained on the METLIN SMRT dataset, enhances the system by accurately identifying tampered or compromised reagents. Additionally, a QR-based kill switch prevents the reuse of packaging, effectively eliminating counterfeit risks. With low transaction costs, high verification reliability, and no requirement for additional hardware or changes in logistics workflows, BioToken offers a scalable, secure, and industry-ready solution for ensuring trust, transparency, and quality in biochemical supply chains.

Keywords- ***Blockchain, zero-knowledge proofs, NFT, supply chain, reagent authentication, anomaly detection, ZK-SNARKs, HPLC, Polygon, ERC-721***

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Chapter 1: Introduction

1.1 Introduction to the Project

The global biotechnology and pharmaceutical supply chain operates on a "high-trust, low-verification" model that has become increasingly inadequate in the face of modern threats. Biochemical reagents - ranging from CRISPR-Cas9 nucleases and monoclonal antibodies to PCR master mixes and specialized enzymes - form the foundational inputs of life sciences research, clinical diagnostics, and therapeutic development. The integrity of these substances is not merely a quality concern; it is directly linked to patient safety, scientific reproducibility, and the credibility of the research enterprise itself.

Historically, the primary method of reagent authentication has been the Certificate of Analysis (COA), a static PDF or paper document issued by the manufacturer at the time of production. While this instrument served adequately in localized, low-volume supply chains, it is fundamentally insufficient for the globalized biotech market of today. COAs are trivially forgeable, offer no real-time visibility into a reagent's condition post-manufacture, and provide no mechanism to detect degradation caused by cold-chain failures during international transit. These failures are often silent - a temperature excursion may chemically alter a reagent without any visible change to its packaging or appearance, meaning the damage is only discovered when an experiment fails, often after months of work and hundreds of thousands of dollars in investment.

This project introduces **BioToken**, a decentralized, end-to-end verification framework that transforms high-value biochemical reagents into Programmable Digital Assets. BioToken combines three powerful technologies: Blockchain (specifically the Polygon Proof-of-Stake network), Non-Fungible Tokens (ERC-721 standard), and Zero-Knowledge Proofs (ZK-SNARKs compiled using Circom). Together, these form an immutable chain of custody from the point of manufacture to the moment of consumption in the laboratory.

The core innovation of BioToken lies in its ZK-AI Validation layer. A manufacturer's proprietary chemical fingerprint - the HPLC chromatogram that uniquely identifies a reagent batch - is processed through a Zero-Knowledge circuit to produce a Verification Key (vk). This vk is a mathematical commitment to the reagent's identity without exposing the underlying data. When a laboratory receives the reagent, a locally-run AI model compares the lab's own scan against the manufacturer's privately shared Gold Standard. If the scans match within a defined tolerance, the AI generates a Zero-Knowledge Proof (π), which is submitted to a smart contract on-chain. The contract verifies π against the stored vk, and if the math holds, the reagent is marked as VERIFIED - with zero raw data ever leaving either party's private environment.

The system also addresses physical tampering through a cryptographic Kill Switch: a frangible QR seal that physically self-destructs when the vial is opened. A single "Consume" transaction then marks the NFT as CONSUMED on-chain, rendering the empty packaging worthless to counterfeiters. BioToken thus creates a complete, tamper-proof digital lifecycle - MINTED → IN_TRANSIT → RECEIVED → VERIFIED → CONSUMED - for every reagent batch it tracks.

1.2 Objectives

The primary aim of this project is to design and implement a functional, decentralized reagent provenance and verification framework that eliminates blind trust from the biotech supply chain without requiring manufacturers to expose proprietary chemical data.

The specific objectives of the BioToken project are as follows:

Objective 1: To develop an ERC-721 smart contract on the Polygon Amoy testnet capable of minting reagent NFTs, recording custody events, executing on-chain ZK-proof verification, and implementing a cryptographic Kill Switch via the consumeToken() function.

Objective 2: To design and compile a Zero-Knowledge circuit using Circom/snarkjs that accepts HPLC chromatographic data as a secret witness and generates a Verification Key (vk) and corresponding Zero-Knowledge Proof (π), enabling private verification of reagent authenticity.

Objective 3: To train an AI anomaly detection model on publicly available NIR/HPLC spectral datasets that can detect molecular degradation or adulteration in received reagent samples at a sensitivity threshold of 2–5% peak deviation, with greater than 90% classification accuracy.

Objective 4: To build a backend API using FastAPI that handles courier webhook-based custody logging, a cross-company consensus validator that rejects outlier fingerprints prior to NFT minting, and an encrypted peer-to-peer channel for Gold Standard delivery.

Objective 5: To develop a dual-portal decentralized application (DApp) in React.js -a Manufacturer Minting Dashboard and a Lab Verification Scanner -that provides an accessible, user-friendly interface for all four stages of the BioToken lifecycle.

1.3 Motivation

The motivation for BioToken emerges from a convergence of documented real-world failures and a critical gap in existing technological solutions.

The Financial and Scientific Cost of Counterfeit Reagents: Substandard and counterfeit reagents have been identified as a root cause of 36.1% of all irreproducible research results, costing the United States alone an estimated $28 billion annually. A single failed ELISA reagent has been documented to waste up to two years of research time and $500,000 in funding. These are not abstract statistics -they represent careers derailed, clinical trials delayed, and medical breakthroughs deferred.

The Gap Between Existing Systems and Real Needs: A thorough review of existing literature revealed that while several blockchain-based pharmaceutical traceability systems have been proposed, none adequately address the specific challenges of biochemical research

reagents. Systems such as PharmaCrypt and PrivChain focus on drug packaging location rather than the chemical integrity of the substance inside. None incorporate an AI-based anomaly detection layer, and none provide a privacy-preserving mechanism that allows verification without exposing manufacturer trade secrets.

Field Validation from ICMR: The team conducted a consultation visit to the National Institute for Research in Reproductive and Child Health (ICMR-NIRRCH), Mumbai, where Dr. Prashant Warang confirmed that no automated or AI-based system currently exists for verifying reagent quality at the point of receipt. Quality checks remain manual, relying on visual inspection and basic temperature logs -methods wholly insufficient for detecting molecular-level degradation. His endorsement of the BioToken concept as novel and practically valuable provided direct domain validation for this project's direction.

The Unique Technical Opportunity: The convergence of mature ZK-SNARK tooling (Circom, snarkjs), energy-efficient blockchain infrastructure (Polygon PoS), accessible AI/ML frameworks (scikit-learn, PyTorch), and large publicly available spectral datasets (METLIN-SMRT, RepoRT) means that a full working prototype of this system is realistically achievable within a single academic semester. BioToken sits at the precise intersection of cryptography, artificial intelligence, and life sciences -a combination that no prior system has unified into a single coherent framework.

1.4 Brief Introduction to Literature

A systematic review of existing research and deployed systems in the domains of pharmaceutical supply chain security, blockchain-based traceability, Zero-Knowledge cryptography, and AI-driven spectral analysis was conducted to establish the technical foundation for BioToken.

Several studies have explored blockchain-based anti-counterfeiting systems for pharmaceutical supply chains. Singh et al. (2022) and Jain and Somani (2022) proposed blockchain frameworks for tracking COVID-19 vaccines and pharmaceutical products using smart contracts, demonstrating the viability of distributed ledger technology for supply chain verification. However, both systems were designed for packaged drug products and relied on barcode scanning at distribution nodes -tracking the location of the packaging rather than verifying the chemical integrity of the substance within.

Badhotiya et al. (2020) proposed PharmaCrypt, a blockchain-based tool for counterfeit drug detection using barcode scanning at each supply chain handoff. While effective at location tracking, the system exposed raw batch data on-chain, creating a trade secret leakage vulnerability and offering no protection against post-packaging chemical tampering.

In the domain of privacy-preserving provenance, research on ZKP-based supply chain frameworks (such as PrivChain, 2022) demonstrated the theoretical applicability of Zero-Knowledge Range Proofs to hide sensitive supply chain data while preserving traceability on Hyperledger Fabric. However, these systems lacked any physical tamper-evidence mechanism and contained no AI component capable of detecting degradation or adulteration at the molecular level.

On the AI and analytical chemistry side, Samuelsson et al. (2025) and Rahal et al. (2025) demonstrated that machine learning models -including Gradient Boosting, Random Forest, and 1D-CNNs -can achieve high accuracy in classifying and predicting chromatographic retention times in HPLC datasets. These studies confirmed the technical feasibility of training

AI models on spectral data for quality assessment, though neither study connected this capability to any blockchain or provenance verification system.

The literature review identified a clear and consistent gap: no prior system unifies chemical-level AI verification, Zero-Knowledge privacy-preserving cryptography, NFT-based digital asset tracking, and a physical tamper-evidence mechanism into a single end-to-end framework. BioToken is designed to fill this gap.

1.5 Problem Definition

The global supply chain for high-value biochemical reagents is currently afflicted by a fundamental insecurity of provenance, rooted in three interconnected failures.

The Trust Gap and Data Asymmetry: Manufacturers possess the definitive chemical fingerprint of each reagent batch, typically in the form of HPLC or Mass Spectrometry chromatograms. Because this data is proprietary and commercially sensitive, it is never shared directly with the purchasing laboratory. Instead, manufacturers provide a simplified Certificate of Analysis (COA) -a document that is static, easily forged, and provides no mechanism for real-time, independent verification. This data asymmetry means that laboratories are structurally forced to rely on the manufacturer's word, with no cryptographic or scientific means of confirming that the liquid in the received vial matches the substance that was tested at the factory.

Empty Bottle Fraud: Authentic, branded reagent containers -which may cost hundreds to thousands of dollars each -represent high-value targets for fraudsters. Empty vials scavenged from laboratory waste can be refilled with diluted, expired, or entirely different substances and reintroduced into the supply chain. Because the packaging is genuine, neither visual inspection nor conventional barcode scanning can detect the substitution. Current tracking systems trace the location of the container, not the identity or consumption state of its contents.

Cold-Chain Degradation Without Detection: Biological materials are highly sensitive to thermal fluctuations. A temperature excursion during air freight cargo clearance or last-mile delivery can denature enzymes, degrade antibodies, or reduce PCR efficiency -without any visible change to the vial or its seal. Standard logistics temperature logs are centralized and can be retroactively altered, providing no immutable record of cold-chain compliance. The damage is typically discovered only when experiments fail, by which point the chain of accountability has dissolved.

Together, these three failure modes create a systemic vulnerability that costs the global research community billions of dollars annually and poses direct risks to patient safety in clinical and therapeutic settings. The current system has no mechanism to detect any of these failures in real time, before the compromised reagent is consumed in an irreversible experiment.

1.6 Scope of the Work

BioToken focuses specifically on the lifecycle management and authenticity verification of high-value liquid biochemical reagents used in pharmaceutical manufacturing and research laboratory settings. The scope encompasses four clearly defined functional stages.

The system covers: the generation and on-chain anchoring of cryptographic chemical fingerprints at the point of manufacture; the automated recording of logistics custody events via webhook integration with existing courier management software; the local AI-based chemical scan comparison and ZK-proof generation at the receiving laboratory; and the cryptographic consumption event that marks a reagent as irreversibly used.

The target users of BioToken are pharmaceutical manufacturers who produce high-value reagent batches, research and clinical laboratories that purchase and consume such reagents, and logistics intermediaries whose existing software can be integrated via webhook APIs.

The system is scoped for deployment on the Polygon Amoy testnet for this project phase, with all AI model training conducted on publicly available NIR and HPLC spectral datasets as proxies for proprietary reagent data.

The following are explicitly outside the current scope: IoT sensor hardware for real-time temperature monitoring during transit (a noted future enhancement), mainstream pharmaceutical drug tracking for patient-dispensed medicines (a distinct regulatory domain), integration with regulated central pharmaceutical authority databases, and deployment on Polygon mainnet (reserved for post-prototype commercialization). The consensus validator is also designed with a known limitation for reagents supplied by a single vendor, requiring manual auditor sign-off in such cases.

1.7 Feasibility Study

Technical Feasibility: All core technologies required for BioToken are mature, open-source, and well-documented. The Polygon Amoy testnet provides a free, low-latency blockchain environment for smart contract deployment. Circom and snarkjs are production-grade ZK-SNARK tools with an established developer community. Solidity is the industry-standard language for EVM-compatible smart contracts. Python's scikit-learn and PyTorch libraries are widely used for spectral data classification. React.js and FastAPI are standard frameworks for frontend and backend development respectively. IPFS via Pinata is a reliable decentralized storage solution. No specialized hardware beyond a standard developer laptop (16GB RAM) is required for ZK proof generation or AI model training during this project phase. Publicly available datasets -the METLIN-SMRT database and the RepoRT chromatography dataset -provide over 7,000 NIR spectra and 25,000 HPLC traces for model training.

Cost Feasibility: The project is designed to operate at near-zero direct cost during the development phase. Polygon Amoy testnet transactions are free. Pinata's IPFS service offers a free tier sufficient for prototype metadata storage. All frameworks and tools are open-source. The only anticipated cost in a production deployment would be Polygon mainnet gas fees, which are targeted at under $0.20 per verification transaction -significantly lower than the cost of manual lab auditing or third-party reagent re-testing. This makes BioToken economically competitive with existing verification methods even at commercial scale.

Operational Feasibility: The system is designed with minimizing workflow disruption as a core principle. Couriers are not required to interact with any new software -custody events are logged automatically via webhook integration with their existing delivery management systems. Laboratory scientists perform only two actions: a single QR scan on receipt and a single "Consume" tap when opening the vial. Manufacturers interact through a dashboard UI for batch minting. No blockchain wallet management or cryptographic key handling is exposed to end users.

Social Feasibility: The project was validated through a direct consultation with domain experts at ICMR-NIRRCH, Mumbai, where researchers confirmed the reality of the problems BioToken addresses and endorsed the proposed approach. The system does not require any change in laboratory scientific practice, only in the administrative verification step. Its adoption benefits all stakeholders: manufacturers gain protection for their intellectual property, laboratories gain verifiable authenticity guarantees, and the broader scientific community gains a tool that directly supports research reproducibility.

1.8 SDG Mapped, Societal Impact and Benefits

BioToken is mapped to two United Nations Sustainable Development Goals.

SDG 9 - Industry, Innovation and Infrastructure: BioToken introduces a novel cryptographic-AI framework that advances the state of the art in decentralized supply chain security. By demonstrating the feasibility of combining Zero-Knowledge Proofs with AI-based spectral analysis for physical asset verification, the project contributes to building resilient, innovation-driven infrastructure for the life sciences industry.

SDG 12 - Responsible Consumption and Production: BioToken directly promotes responsible production by creating an immutable record of reagent authenticity and lifecycle status. The Kill Switch mechanism ensures that consumed packaging cannot re-enter the supply chain as counterfeit, directly reducing waste and fraudulent re-use. The system also discourages the production and distribution of substandard reagents by making detection mathematically certain rather than probabilistically likely.

Beyond SDG alignment, BioToken carries significant societal benefits. In clinical settings, preventing the use of degraded or counterfeit reagents in diagnostic assays directly protects patient safety - a false negative produced by a failed ELISA reagent can have life-threatening consequences. In research settings, ensuring reagent integrity supports the reproducibility of scientific results, a foundational requirement of evidence-based medicine and drug development. The financial savings from reduced irreproducible research -estimated at tens of billions of dollars annually at the global scale -represent resources that can be redirected toward genuine scientific progress. By open-sourcing all code, ZK circuits, and AI model weights on GitHub, the project also ensures that the framework remains accessible to underfunded research institutions that are currently most vulnerable to reagent fraud.

1.9 Organization of the Report

This report is structured across six chapters as follows.

Chapter 1 - Introduction provides an overview of the BioToken project, its context within the global biotech supply chain, the objectives and motivation behind the work, a brief survey of relevant literature, a formal problem definition, the scope of the system, a feasibility analysis across technical, economic, operational, and social dimensions, and the SDG alignment and societal impact of the proposed solution.

Chapter 2 - Literature Survey presents a detailed review of a minimum of five research papers from related domains including blockchain-based pharmaceutical traceability, Zero-Knowledge proof systems for supply chains, AI-driven spectral anomaly detection, and NFT-based digital asset frameworks. Each paper is analyzed for its key contributions, methodological approach, and research gaps that BioToken addresses.

Chapter 3 - Design and Implementation describes the complete system architecture of BioToken, including functional and non-functional requirements, the proposed ZK circuit design and algorithm, UML diagrams (use case, sequence, and class diagrams), a block diagram of the Physical-to-Digital-to-Physical loop, a Gantt chart for project planning, cost estimation, and detailed hardware and software requirements.

Chapter 4 - Proposed Results and Discussions covers the dataset collection and preprocessing methodology, the experimental setup for AI model training and ZK circuit compilation, the evaluation metrics defined for the project (ZK verification success rate, AI sensitivity, gas fee per transaction, proof generation latency, Kill Switch replay rejection rate), and a comparative analysis of BioToken against existing systems.

Chapter 5 - Plan of Action for the Next Semester summarizes all work completed to date including smart contract development, ZK circuit compilation, AI model training progress, and backend API construction, and presents the detailed plan of action for the second semester, including frontend completion, full system integration, stress testing, and final evaluation.

Chapter 6 - Conclusion summarizes the contributions of the BioToken project, reflects on its limitations and known constraints, and proposes directions for future work including IoT cold-chain sensor integration, Polygon mainnet deployment, and extension of the framework to additional high-value asset classes beyond biochemical reagents.

Chapter 2: Literature Survey

2.1 Introduction

The integrity of biochemical reagents is critical as precision medicine and synthetic biology scale. Current provenance systems face vulnerabilities that span blockchain management, Zero-Knowledge (ZK) cryptography, and analytical chemistry. This chapter reviews six pivotal papers to establish the state of the art in blockchain traceability, ZKP privacy, and AI-driven chromatographic analysis, identifying the boundaries BioToken aims to transcend.

2.2 Related Work

- **Paper 1:** ***Counterfeit Drugs in the Indian Market*** **(2026):** Establishes that pharmaceutical counterfeiting is a systemic issue in fragmented supply chains. However, it focuses on finished drugs (pills/injectables) rather than high-value research reagents like CRISPR components and lacks any physical tamper-evidence mechanisms.
- **Paper 2:** ***PrivChain: Privacy in Blockchain Supply Chains*** **(2022):** Uses ZK Range Proofs for data privacy. While it proves the viability of ZKPs in logistics, it remains purely digital. It cannot verify the physical substance's integrity or prevent "empty bottle" fraud, and its incentive model relies on impractical cryptocurrency structures.
- **Paper 3:** ***Blockchain for COVID-19 Vaccine Traceability*** **(2022):** Demonstrates effective smart contract-based custody logging. However, it tracks packaging rather than molecular content and lacks AI for detecting chemical degradation—a vital requirement for volatile biochemicals.
- **Paper 4:** ***PharmaCrypt: Blockchain to Counterfeit Drugs*** **(2020):** Uses barcodes for handoff verification but stores raw batch data on-chain. This exposes manufacturer trade secrets. It also fails to verify if the substance inside the packaging is authentic or adulterated.
- **Paper 5:** ***ML for Oligonucleotide Retention Time Prediction*** **(2025):** Confirms that ML can accurately process HPLC (High-Performance Liquid Chromatography) data. However, the models are research tools for prediction, not security tools for authenticity or fraud detection.
- **Paper 6:** ***ML Anomaly Detection in Cloud Labs*** **(2025):** Achieves high accuracy (0.96) in detecting HPLC hardware anomalies (e.g., air bubbles). While it proves ML's efficacy on spectral data, it does not address reagent identity or integrate with supply chain provenance.

2.3 Comparative Analysis

The following table maps existing literature against the nine dimensions addressed by BioToken:

Table 2.1: Comparative Analysis of BioToken vs Existing Systems

Feature / Dimension	**PharmaCrypt**	**PrivChain**	**HPLC ML**	**BioToken**
Blockchain Custody Tracking	✓	✓	✖	✓
Zero-Knowledge Privacy	✖	✓	✖	✓
Chemical/Molecular Verification	✖	✖	✓ (Partial)	✓
AI Anomaly Detection	✖	✖	✓	✓
Physical Tamper-Evidence	✖	✖	✖	✓
Trade Secret Protection	✖	✓	N/A	✓
Offline/Local Deployment	✓	✓	✖	✓

2.4 Research Gaps

1. **Absence of Chemical Verification:** Existing blockchains track "the box," not "the substance." BioToken bridges this by linking chromatographic analysis to on-chain records.
2. **Trade Secret Exposure:** Current systems often require sharing sensitive formulation data. BioToken uses ZK-SNARKs to verify chemical fingerprints without exposing raw data.
3. **Physical-Digital Disconnect:** No existing work prevents the reuse of authentic vials for fake reagents. BioToken's **QR Kill Switch** ensures the digital token is destroyed upon opening.
4. **Lack of Authenticity AI:** While AI exists for chemistry, it hasn't been used for binary "Genuine vs. Adulterated" classification in a supply chain context.
5. **Lack of Integrated Frameworks:** Consultations with **ICMR-NIRRCH** confirm that practitioners lack any point-of-receipt automated verification. BioToken is the first end-to-end system grounded in these real-world operational constraints.

Chapter 3: Design Implementation

3.1 Introduction

The design implementation phase transitions the conceptual framework of the BioToken system into a concrete, technical architecture. This section outlines the functional and nonfunctional requirements, proposed algorithms, and structural diagrams necessary to build a decentralized reagent provenance system. By integrating Zero-Knowledge Proofs (ZK-SNARKs), Machine Learning (XGBoost/Gradient Boosting), and Account Abstraction (Privy and Pimlico), the proposed implementation completely abstracts Web3 complexities while mathematically guaranteeing supply chain integrity without exposing proprietary trade secrets.

3.2 Nonfunctional and Functional Requirement

Requirement Gathering
To ensure BioToken effectively bridges the "Trust Gap" in the biotech supply chain, the system requirements were gathered based on the limitations of existing track-and-trace solutions, which only secure packaging data rather than chemical integrity.

Functional Requirements (What the system must do):

- **Digital Birth & Minting:** The system must allow manufacturers to generate a Zero-Knowledge Verification Key (vk) from HPLC retention times and mint an ERC-721 NFT containing this key.
- **Consensus Validation:** The backend must cross-reference a manufacturer's submitted vk against a marketplace consensus range before approving the mint.
- **Automated Custody Logging:** The system must receive webhooks from existing courier software to update the NFT status (e.g., IN_TRANSIT, RECEIVED) without manual wallet signing.
- **AI-Driven Anomaly Detection:** The local lab client must utilize a machine learning model (trained on the METLIN-SMRT dataset) to predict acceptable retention times and detect >2-5% chemical degradation.
- **ZK-Proof Generation:** Upon a successful AI match, the local client must generate a Groth16 ZK-Proof (π) asserting chemical authenticity.
- **Gasless On-Chain Verification:** The system must utilize Privy for user authentication and Pimlico as a Paymaster to submit the ZK-Proof to the smart contract gaslessly.
- **Cryptographic Kill Switch:** The system must allow a lab scientist to trigger a consumeToken() function upon opening the physical vial, permanently burning the NFT to prevent Empty Bottle Fraud.

Nonfunctional Requirements (How the system must behave):

- **Data Privacy (Zero-Knowledge):** Raw chemical formulation data must never leave the manufacturer's or lab's local environments.
- **Usability (Account Abstraction):** Lab scientists must be able to interact with the blockchain using standard email logins (Privy) without needing to manage private keys or pay gas fees.
- **Performance:** ZK-Proof generation must occur locally in under 60 seconds on standard laboratory hardware.
- **Immutability:** Once a reagent's status changes to CONSUMED, the smart contract must strictly revert any future attempts to verify or transfer it.

3.3 Proposed Design and Algorithm

Proposed Designs and Required Algorithm Discussion

BioToken's core novelty lies in the ZK-AI Verification Algorithm (Stage 3). Rather than relying on simple deterministic smart contracts—which would falsely reject authentic chemicals due to minor hardware calibration differences across labs—BioToken utilizes an AI-driven method-independent pipeline.

Algorithm: AI-Assisted ZK Proof Trigger

1. **Input:** Lab scans the received reagent, extracting a numerical feature vector (molecular weight, logP, measured Retention Time).
2. **Prediction:** The local XGBoost model, trained on the METLIN-SMRT dataset, predicts the expected Retention Time (RT) based on the specific laboratory hardware.
3. **Three-Way Comparison (Anomaly Detection):**
    a. Check A: Is the Measured RT within the acceptable tolerance of the Manufacturer's Gold Standard?
    b. Check B: Is the Measured RT within the expected RMSE of the AI's predicted RT?
4. **Decision:**
    a. If False: Flag anomaly, mark NFT as SUSPICIOUS, halt pipeline.
    b. If True: Pass the measured RT as a hidden witness into the Circom circuit.
5. **Proof Generation:** The circuit computes the Groth16 proof (π) and forwards it to the Pimlico Paymaster for on-chain verification against the public vk.

3.4 Architectural Diagrams

3.4.1 UML Diagrams

Phase 1 - Digital Birth & Consensus

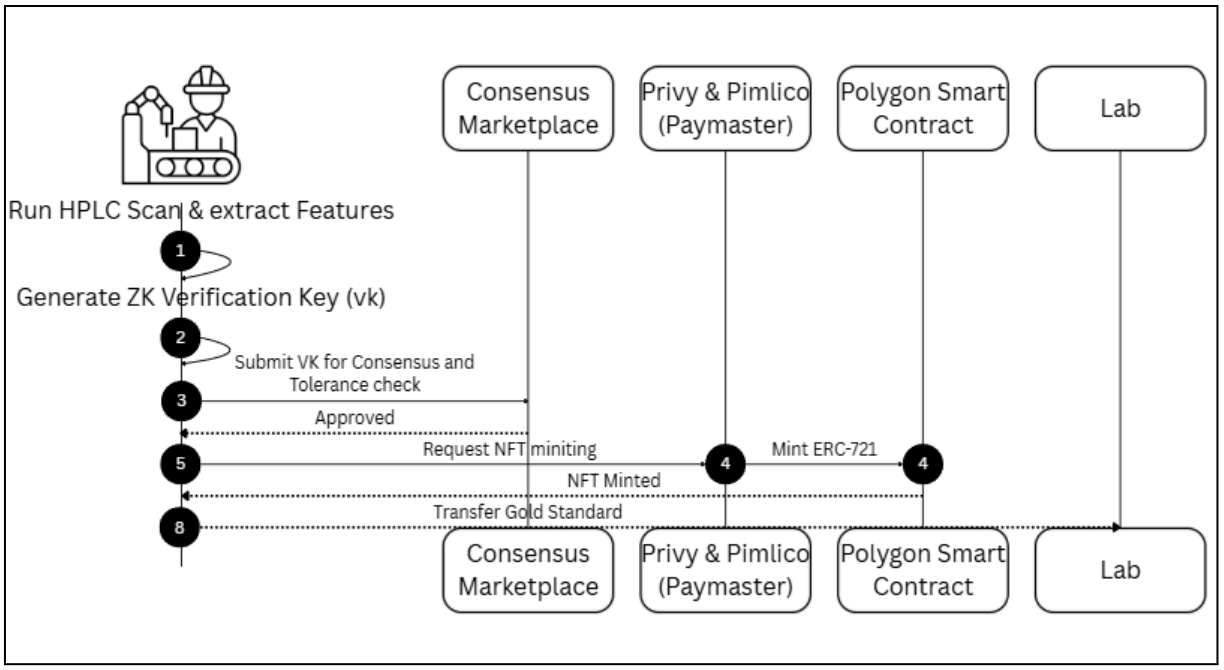


Figure 3.1: Phase 1 - Digital Birth & Consensus Sequence Diagram

This diagram illustrates how a physical reagent is converted into a verifiable digital asset (NFT) without exposing the manufacturer's trade secrets.

- **Data Extraction & ZK Key (Steps 1-2):** The Manufacturer performs an HPLC chemical scan and generates a Zero-Knowledge Verification Key (vk) locally. The secret formula is never exposed.
- **Consensus Validation (Step 3):** The vk is checked by the Consensus Marketplace to ensure the chemical signature falls within an acceptable industry tolerance range, preventing fraudulent baselines.
- **Gasless NFT Minting (Steps 4-7):** Once approved, the system uses Privy and Pimlico (Paymaster) to sponsor gas fees and automatically mint an ERC-721 NFT on the Polygon blockchain containing the vk.
- **Secure Handoff (Step 8):** The raw "Gold Standard" data is securely transferred entirely off-chain to the purchasing Lab so they can verify the physical vial when it arrives.

Phase 2 - Automated Custody Chain

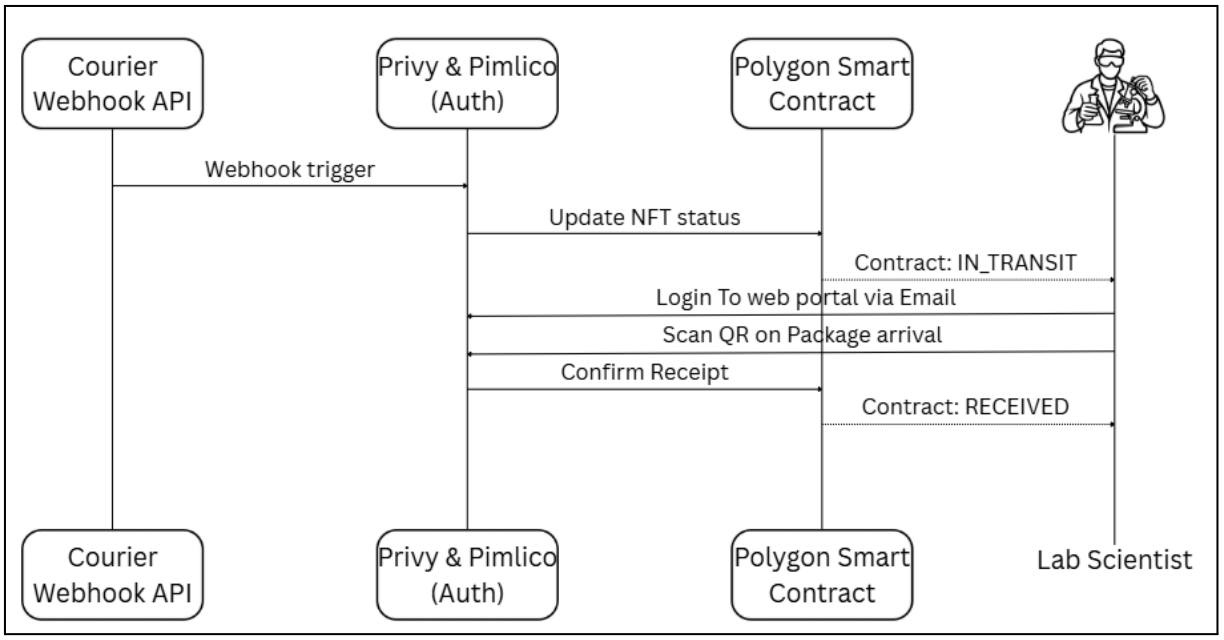


Figure 3.2: Phase 2 - Automated Custody Chain Sequence Diagram

This diagram illustrates how the system tracks the physical reagent during transit using automated logs, requiring no Web3 knowledge from the logistics providers or the receiving scientists.

- **Automated Transit Logging:** The Courier's existing delivery software sends a standard webhook trigger. This automatically updates the NFT status to "IN_TRANSIT" on the Polygon Smart Contract without requiring the courier to use a crypto wallet.
- **Frictionless Login:** The Lab Scientist accesses the BioToken web portal using a simple email login powered by Privy, bypassing the need to manage complex Web3 seed phrases.
- **Receipt Confirmation:** Upon physical arrival of the package, the Lab Scientist simply scans the QR code. The system automatically confirms receipt and updates the smart contract status to "RECEIVED".

Phase 3 - ZK-AI Verification & Kill Switch

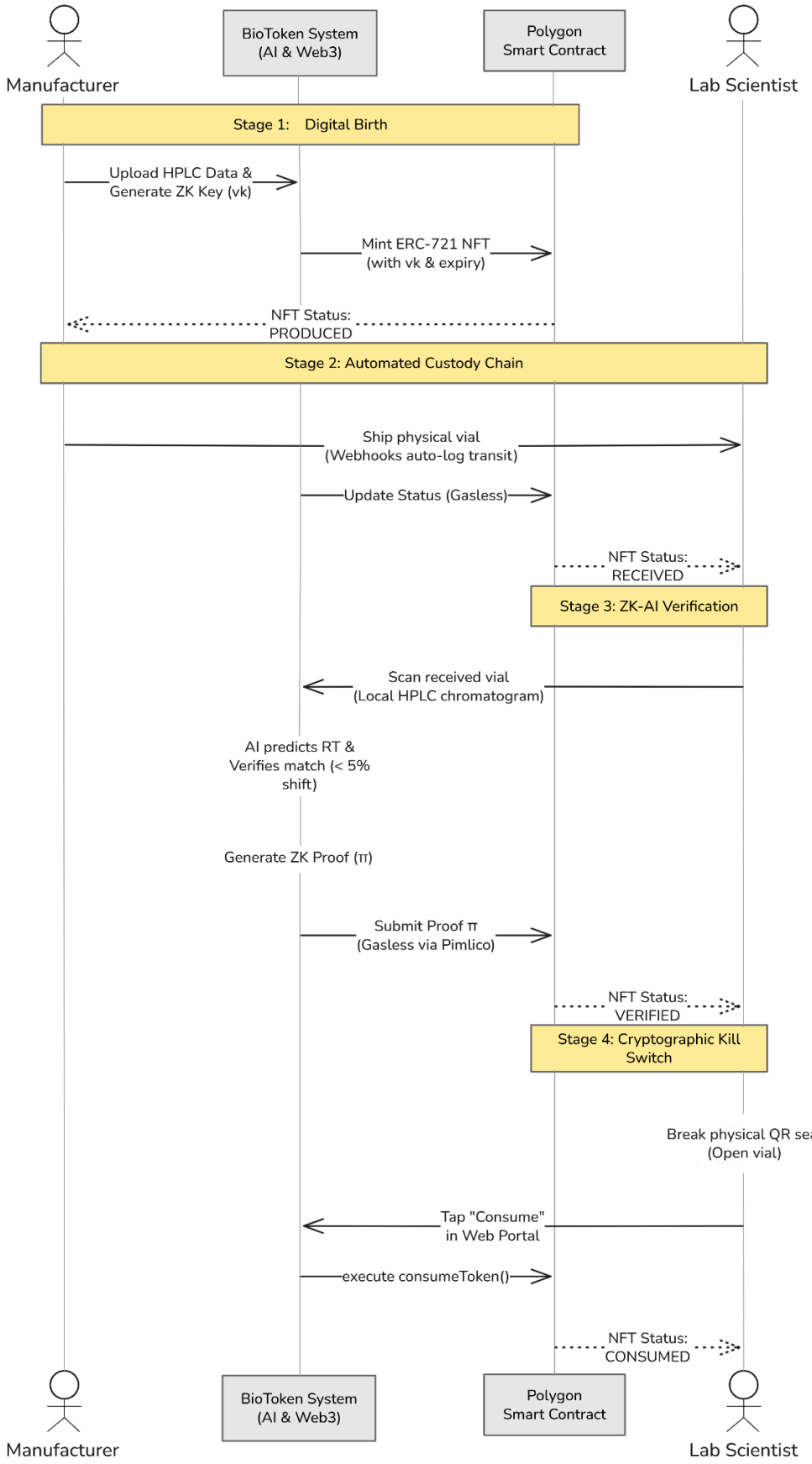


Figure 3.3: Phase 3 - ZK-AI Verification & Kill Switch Sequence Diagram

This sequence diagram illustrates the complete end-to-end lifecycle of a reagent, emphasizing the AI-driven verification and anti-counterfeiting mechanisms at the laboratory.

- **Digital Birth & Transit (Stages 1-2):** The Manufacturer converts chemical data into a ZK Key (vk) and mints the reagent's NFT. Automated webhooks track the package seamlessly until it arrives at the Lab.
- **AI-Driven Verification (Stage 3):** The Lab Scientist scans the physical vial locally. The AI engine checks for degradation (detecting shifts > 5%). If the chemical is authentic, it generates a Zero-Knowledge Proof (π).
- **Gasless On-Chain Proof (Stage 3):** The ZK Proof (π) is submitted to the Polygon smart contract via a Paymaster, costing the lab scientist zero gas fees. The blockchain updates the status to "VERIFIED".
- **Cryptographic Kill Switch (Stage 4):** To use the reagent, the scientist physically breaks the frangible QR seal on the vial and taps "Consume" in the web portal. The smart contract executes consumeToken(), permanently updating the NFT to "CONSUMED" and destroying any resale value for counterfeiters.

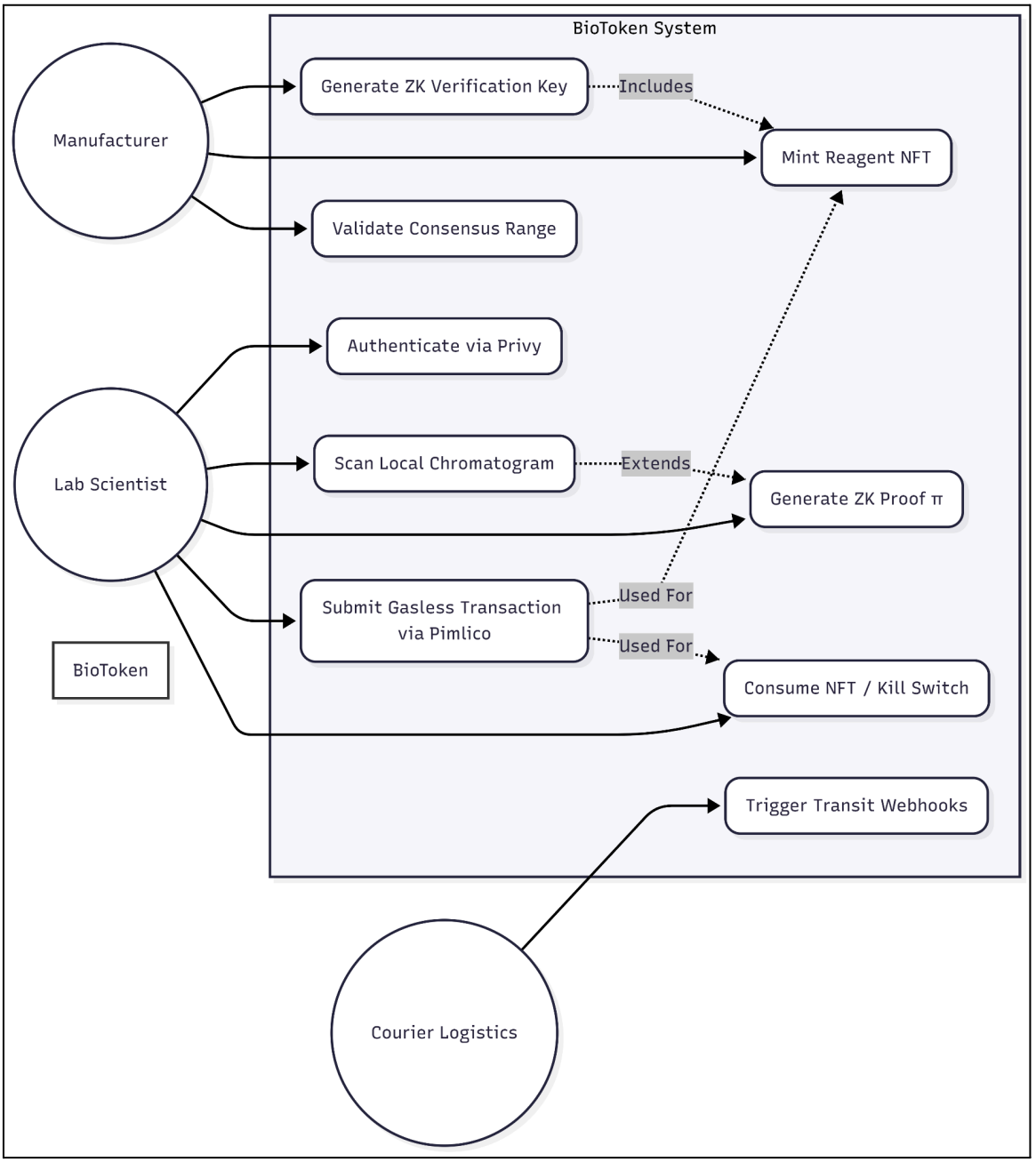


Figure 3.4: Use Case Diagram - BioToken System Actors and Functions

This use case diagram maps out the primary users (Actors) of the BioToken platform and the specific actions (Use Cases) they are authorized to perform within the system boundary.

- **Manufacturer:** Responsible for the reagent's digital inception. They generate the Zero-Knowledge Verification Key, validate it against the marketplace consensus, and execute the minting of the ERC-721 NFT.
- **Courier Logistics:** Their existing logistics software triggers automated webhooks to update the package's transit status, requiring no manual blockchain interaction.
- **Lab Scientist:** The primary end-user. They securely authenticate via Privy, scan the local chromatogram (which generates the ZK Proof if authentic), and trigger the physical Kill Switch.
- **Account Abstraction (Pimlico/Privy):** The diagram highlights how gasless transaction submissions are "used for" core smart contract interactions (minting and consuming NFTs), ensuring users never have to manage cryptocurrency wallets.

3.4.2 Block diagrams

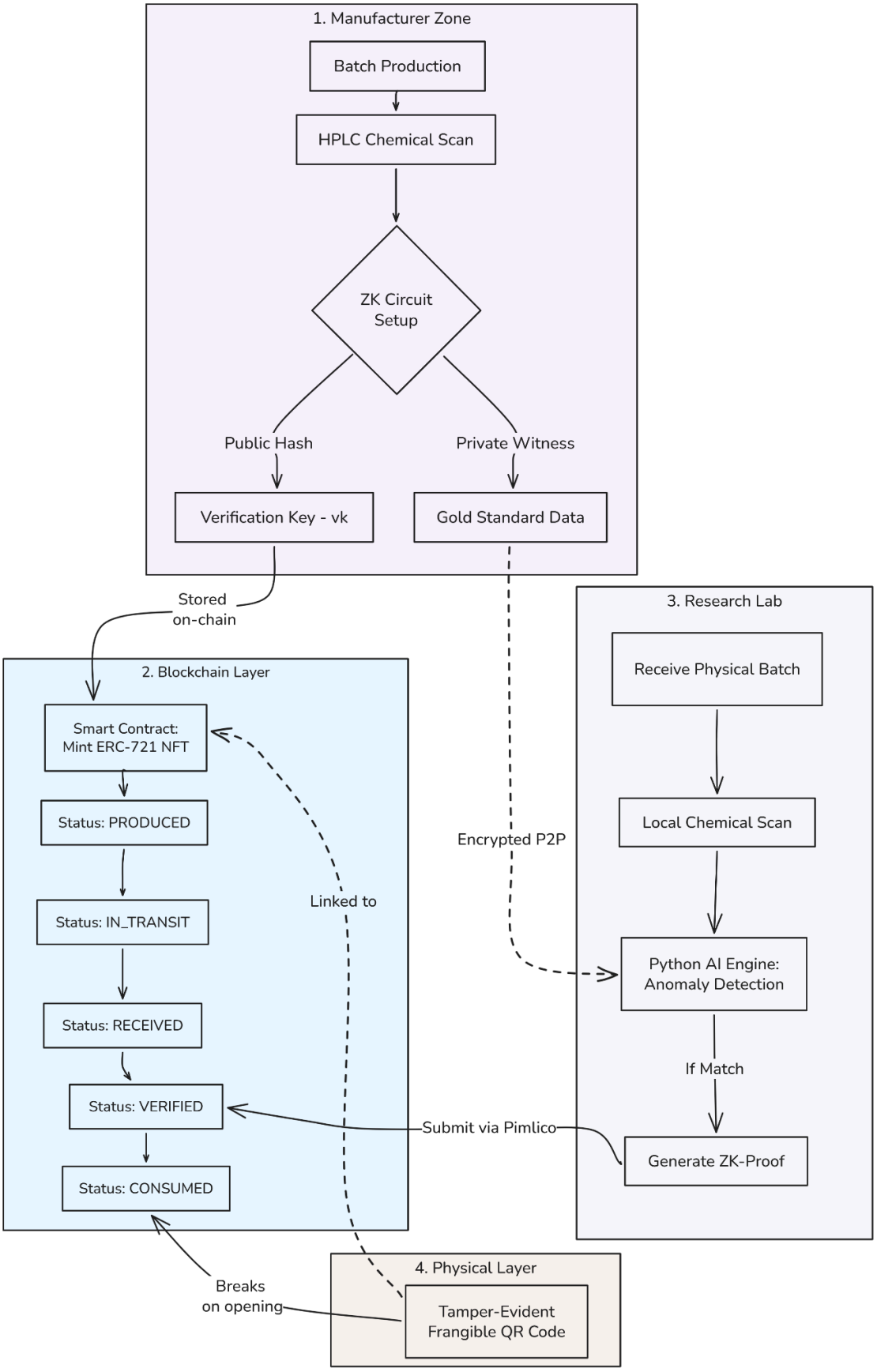


Figure 3.5: System Architecture Block Diagram

The diagram illustrates the four-layer BioToken pipeline. At the manufacturer stage (top), a batch undergoes HPLC chemical scanning, feeding a ZK Circuit Setup that bifurcates into two outputs: a public Verification Key (vk) stored on-chain, and private Gold Standard Data transmitted to the lab via an encrypted P2P channel.

The Blockchain Layer (left) captures the full NFT lifecycle through five sequential states — PRODUCED → IN_TRANSIT → RECEIVED → VERIFIED → CONSUMED — managed by an ERC-721 smart contract. The Research Lab Layer (right) represents the lab-side verification flow: the physical batch is received, locally scanned, and passed through the Python AI anomaly detection engine. If the scan matches the Gold Standard, a ZK-Proof is generated and submitted to the smart contract via Pimlico, triggering the VERIFIED state transition. Finally, the Physical Layer (bottom) shows the tamper-evident frangible QR code, which self-destructs on vial opening and triggers the CONSUMED state, preventing replay attacks on spent reagents.

3.5 Proposed Planning

3.5.1 Timeline chart (Gantt chart)

The project execution is divided into four distinct phases over 16 weeks.

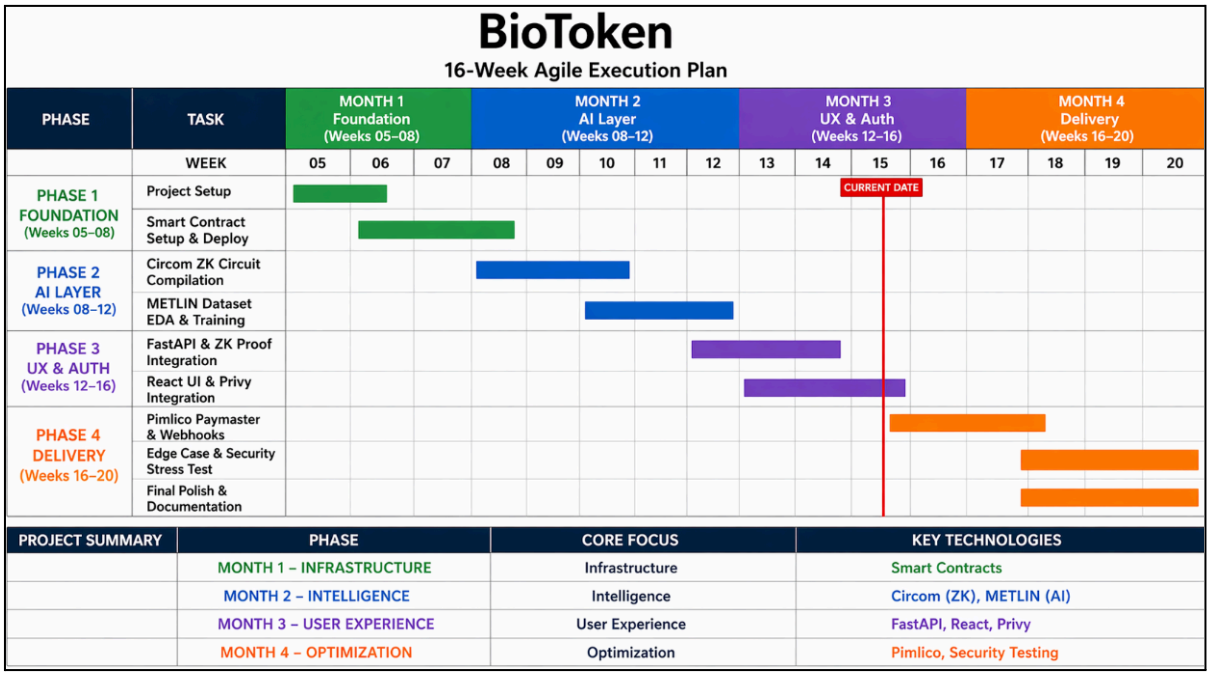


Figure 3.6: BioToken 16-Week Agile Execution Timeline

This Gantt chart outlines the 16-week agile development lifecycle of the BioToken project, structured across four monthly phases. The red vertical line indicates the current progress, showing the project is successfully transitioning into its final delivery phase.

- **Month 1 - Foundation (Infrastructure):** Dedicated to establishing the core Web3 backend, including environment setup and deploying the ERC-721 smart contracts on the Polygon testnet.
- **Month 2 - AI Layer (Intelligence):** Focuses on compiling the Zero-Knowledge (Circom) circuits and training the XGBoost anomaly detection model using the METLIN dataset.
- **Month 3 - UX & Auth (User Experience):** Bridges the system components by building the

FastAPI backend, developing the React.js frontend, and integrating Privy for seamless user authentication.

- **Month 4 - Delivery (Optimization):** The final phase is reserved for implementing the Pimlico Paymaster (gasless transactions), rigorous security stress-testing (including the Kill Switch), and finalizing project documentation.

3.5.2 Cost Estimation

The financial viability of the BioToken framework is rooted in its highly optimized Web3 architecture. By utilizing Layer-2 scaling solutions (Polygon), Account Abstraction (Pimlico), and lightweight Machine Learning models (XGBoost), the system minimizes both computational overhead and transaction fees.

1. Capstone Prototype Costs (Current Implementation)

For the 4-month academic development lifecycle, the project strictly leverages free-tier infrastructure, testnets, and open-source tooling, bringing the total financial expenditure to $0.00.

Table 3.1: Lifecycle Gas Cost Breakdown (Per Reagent Vial)

Component	Technology Used	Purpose	Estimated Cost (USD)
Blockchain Ledger	Polygon Amoy Testnet	Smart contract deployment and transaction logging	$0.00
Decentralized Storage	Pinata (IPFS) Free Tier	Storing encrypted NFT metadata and Verification Keys (vk)	$0.00 (Up to 1 GB / 500 pins)
Account Abstraction	Pimlico (Testnet Paymaster)	Sponsoring gasless transactions for lab scientists	$0.00
Authentication	Privy (Starter Tier)	Email/Social login and embedded wallet generation	$0.00 (Up to 10,000 MAUs)
Frontend Hosting	Vercel (Hobby Tier)	Hosting the React.js dual-portal DApp	$0.00
Backend API	Render / Railway (Free Tier)	Hosting FastAPI webhook receivers and AI inference	$0.00
AI Datasets	METLIN-SMRT & RepoRT	Open-source chromatography datasets for ML training	$0.00 (Open Access)
Total Prototype Cost			$0.00

2. Projected Mainnet Production Costs (Enterprise Scale)

If BioToken were to be adopted by a pharmaceutical manufacturer or regulatory body (e.g., ICMR), the system would migrate to the Polygon Proof-of-Stake (PoS) Mainnet.

A critical metric for supply chain viability is the "Cost Per Vial". BioToken requires exactly four blockchain transactions per reagent lifecycle: Minting, Custody Transfer, ZK-Verification, and Consumption (Kill Switch).

Lifecycle Gas Cost Breakdown (Per Reagent Vial):

Table 3.2: Monthly Fixed Infrastructure Costs (Production)

Action	**Smart Contract Function**	**Est. Gas Used**	**Cost per Tx (USD)**
1. Digital Birth	mintToken()	~120,000	~$0.002
2. Custody Update	transferCustody()	~40,000	~$0.0008
3. ZK Verification	verifyProof(a, b, c, input)	~250,000	~$0.005
4. Kill Switch	consumeToken()	~45,000	~$0.0009
Total Lifetime Cost	Full Traceability Cycle	~455,000 Gas	~$0.0087 per vial

Through Pimlico's ERC-4337 Paymaster, the manufacturer sponsors this $0.0087 cost, completely abstracting it from the end-user.

Monthly Fixed Infrastructure Costs (Production):

To support enterprise-grade reliability and SLAs, the infrastructure would transition to paid tiers:

- Pinata IPFS (Picnic Plan): ~$20.00 / month (unlimited pins, custom gateway).
- Privy & Pimlico (Pro Tiers): ~$50.00 / month (scaled API limits for gas sponsorship).
- AWS / GCP Hosting: ~$40.00 / month (EC2 instances for FastAPI backend and XGBoost AI inference; high compute is not required as XGBoost inference executes in under 100ms).
- Total Fixed Cloud Costs: ~$110.00 / month.

3. Return on Investment (ROI) and Economic Feasibility

The economic argument for BioToken is profoundly asymmetrical. High-value biochemical reagents, such as CRISPR-Cas9 enzymes, specialized antibodies, or ELISA kits, range in price from $500 to $5,000+ per batch. Furthermore, as noted in the ICMR field findings, a single degraded reagent can cause months of failed experiments, wasting hundreds of thousands of dollars in research grants.

3.6 Hardware and Software Requirements

Software Requirements

- **Programming Languages:** Python 3.10+, Solidity ^0.8.20, TypeScript/JavaScript.
- **Machine Learning:** Scikit-Learn, XGBoost, Pandas (Trained on open-source METLIN-SMRT / RepoRT datasets).
- **Cryptography & Blockchain:** Circom 2.0, snarkjs (Groth16), Hardhat v2, Ethers.js.
- Infrastructure SDKs: Privy (Wallet Abstraction), Pimlico (ERC-4337 Paymaster), Pinata SDK.
- **Web Frameworks:** React.js (Frontend), FastAPI (Backend).

Hardware Requirements

- The system is designed to be lightweight and accessible to standard laboratories without requiring dedicated cloud processing or enterprise server racks.
- Development Environment: Standard PC/Mac (Multi-core CPU, 8GB+ RAM, 20GB Free Storage).
- Lab Deployment Target: Any standard laboratory computer running Windows or Linux.
- No GPU Dependency: Because the AI model utilizes Gradient Boosting rather than heavy deep neural networks, and ZK-Proof generation utilizes the highly optimized Groth16 protocol, inference and proof generation can run smoothly on standard local CPU hardware.
- External Integration: Relies on output files (.csv/.txt chromatogram arrays) generated by standard HPLC (High-Performance Liquid Chromatography) laboratory hardware

Chapter 4

4.1 Data Collection, Acquisition, Data Preprocessing & Data Interpretation

Data Collection & Acquisition

The AI module of BioToken relies on the METLIN Small Molecule Retention Time (SMRT) dataset, published in Nature Communications (2019) and made publicly available via Figshare (DOI: 10.6084/m9.figshare.8XY). The dataset was downloaded directly from the Figshare repository using the URL: https://figshare.com/ndownloader/files/18130628.

The SMRT dataset was selected because it provides experimentally measured Reversed-Phase High-Performance Liquid Chromatography (RP-HPLC) retention times for a large and chemically diverse set of small molecules -the same analytical technique used in BioToken's physical reagent fingerprinting process. This makes it directly applicable to the problem of detecting RT deviations caused by tampering, degradation, or substitution.

The raw dataset contains **80,038 rows and 3 columns**:

Table 4.1: Raw SMRT Dataset Structure

Column	Type	Description
pubchem	Integer	PubChem compound identifier
rt	Float	Retention time in seconds (target variable)
inchi	String	IUPAC InChI chemical structure representation

Data Preprocessing

Raw data preprocessing involved the following steps:

1. Molecular Parsing via RDKit Since the dataset provides only InChI strings (raw molecular structure representations) and no numerical features, RDKit was used to parse each InChI string into a molecular object. Out of 80,038 molecules, **79,957 were successfully parsed**. 81 molecules failed due to valence errors and were discarded -a loss of approximately 0.1%, which is acceptable.

2. Retention Time Filtering Exploratory analysis revealed that molecules with RT < 200 seconds form a separate population of very polar, early-eluting molecules that skew the model's learning. These **2,056 molecules were removed**, leaving a final dataset of **77,901 molecules** for modelling.

3. Feature Engineering -Physicochemical Descriptors Nine physicochemical descriptors were computed for each molecule using RDKit functions. These features capture properties directly relevant to RP-HPLC retention behavior:

Table 4.2: Physicochemical Descriptors and RT Correlation

Feature	RDKit Function	RT Correlation
logp	MolLogP()	+0.48
aromatic_rings	CalcNumAromaticRings()	+0.37
mol_weight	MolWt()	+0.26
heavy_atom_count	GetNumHeavyAtoms()	+0.22
ring_count	CalcNumRings()	+0.19
hba	CalcNumHBA()	+0.07
tpsa	TPSA()	+0.05
rotatable_bonds	CalcNumRotatableBonds()	+0.04
hbd	CalcNumHBD()	-0.20

4. Feature Engineering -Morgan Fingerprints 128-bit Morgan fingerprints (ECFP -Extended Connectivity Fingerprints) were computed at radius 2 for each molecule. Each bit encodes the presence or absence of a specific substructure pattern within 2 bonds of each atom. Adding these fingerprints improved the RT predictor's R^2 from 0.57 to 0.77 -the single largest improvement in the pipeline.

5. Synthetic Anomaly Generation Since no labelled genuine/anomaly HPLC dataset exists for this specific use case, training data for the anomaly classifier was generated synthetically from the METLIN dataset:

- **Genuine samples (label = 0):** Observed RT = true RT × (1 + uniform noise of ±1%)
- **Anomaly samples (label = 1):** Observed RT = true RT × (1 + shift of 15–40%)

2,395 genuine samples with percentage deviation > 20% were additionally removed to eliminate ambiguous samples caused by the RT predictor's own uncertainty on difficult molecules. The final training set contained **153,407 samples** (75,506 genuine + 77,901 anomaly).

6. Feature Scaling StandardScaler was applied separately to the RT predictor and the anomaly classifier training data. The fitted scaler objects were saved as scaler.pkl and scaler_classifier.pkl for use during inference.

Data Interpretation

Key insights extracted during EDA:

- **RT Distribution:** Mean RT = 790.11s (≈13 minutes), Std Dev = 206.65s, range from 0.30s to 1,471.70s. The distribution has a slight left skew, confirming the need to remove early-eluting outliers.
- **Strongest predictor:** logP (hydrophobicity) had the highest correlation with RT (+0.48), consistent with the known chemistry of RP-HPLC where more hydrophobic

molecules are retained longer on the column.

- **H-bond donors (hbd) negatively correlated** with RT (−0.20), as polar molecules interact more with the aqueous mobile phase and elute earlier.
- **Morgan fingerprints significantly boosted predictive power**, confirming that local substructural patterns around atoms capture information about molecular-column interactions that bulk physicochemical properties alone miss.
- **pct_deviation** (percentage deviation of observed RT from predicted RT) emerged as the most important feature for the anomaly classifier, as it provides a normalized signal independent of the molecule's absolute RT range.

4.2 Experimental Setup

Software Environment

Table 4.3: Software Requirements

Component	**Technology**	**Version/Detail**
Programming Language	Python	3.11
Virtual Environment	venv	Isolated as venv311
Data Manipulation	pandas, numpy	Standard versions
ML Framework	scikit-learn, XGBoost	pip-installed
Cheminformatics	RDKit	pip-installed
Visualization	matplotlib, seaborn	pip-installed
Model Persistence	joblib	pip-installed
Notebook Environment	Jupyter (ipykernel)	Kernel: biotoken311
Blockchain Layer	Solidity (ERC-721), Hardhat	Polygon Amoy Testnet
ZK Framework	Circom, snarkjs	Groth16 protocol
Backend API	FastAPI / Node.js	Webhook + consensus validator
Frontend	React.js	Lab-facing BioToken app
Storage	IPFS via Pinata	Encrypted NFT metadata
Web3 Bridge	Ethers.js / Web3.py	Smart contract interaction

Note: Python 3.14 was attempted initially but caused XGBoost kernel crashes due to compatibility issues. Python 3.11 was confirmed stable for the entire pipeline.

Hardware Configuration

All model training and evaluation was conducted on a standard development laptop with 16 GB RAM. No GPU was required, as XGBoost's CPU parallelism (n_jobs = -1) provided sufficient training speed for the dataset size.

Project Structure

```
biotoken-ai/
├── notebooks/
│   ├── 01_eda.ipynb
│   ├── 02_model.ipynb
│   ├── 03_xgboost.ipynb
│   ├── 04_anomaly.ipynb
│   └── 05_classifier.ipynb
├── models/
│   ├── rt_predictor.pkl
│   ├── scaler.pkl
│   ├── anomaly_classifier.pkl
│   ├── scaler_classifier.pkl
│   └── classifier_info.json
├── src/
│   └── verifier.py
└── data/
    ├── SMRT_dataset.csv
    └── SMRT_features_v2.csv
```

Model Parameters

RT Predictor (XGBoost Regressor):

- n_estimators = 500, learning_rate = 0.05, max_depth = 7
- subsample = 0.8, colsample_bytree = 0.8, random_state = 42
- Train/Test Split: 80/20 → 62,320 train / 15,581 test molecules

Anomaly Classifier (XGBoost Classifier):

- n_estimators = 300, learning_rate = 0.05, max_depth = 6
- subsample = 0.8, colsample_bytree = 0.8, eval_metric = 'auc'
- Decision Threshold: 0.50 (selected by maximizing F1 while keeping FPR < 10%)

4.3 Evaluation Metrics

The BioToken AI module was evaluated using the following metrics, chosen specifically for a binary classification task where both false positives and false negatives carry real operational consequences.

RT Predictor Performance

Table 4.4: RT Predictor Performance Metrics

Metric	**Value**
R^2 (Coefficient of Determination)	0.7706
RMSE (Root Mean Squared Error)	83.49 seconds

R^2 = 0.77 means the model explains 77% of variance in retention times across 77,901

chemically diverse molecules -a strong result for a general-purpose model.

Anomaly Classifier Performance (Final Model -v4)

Table 4.5: Anomaly Classifier Performance Summary

Metric	Value	Target	Status
AUC Score	0.9798	> 0.85	✓ Met
F1 Score	0.9351	> 0.85	✓ Met
Overall Detection Rate	92.0%	> 80%	✓ Met
False Positive Rate	4.9%	< 15%	✓ Met
Decision Threshold	0.50	Optimal F1	✓

Confusion Matrix (Threshold = 0.50)

Table 4.6: Confusion Matrix at Threshold 0.50

	Predicted Genuine	Predicted Anomaly
Actually Genuine	14,363 (TN)	738 (FP)
Actually Anomaly	1,251 (FN)	14,330 (TP)

- **Precision** = TP / (TP + FP) = 14,330 / (14,330 + 738) = **0.951**
- **Recall** = TP / (TP + FN) = 14,330 / (14,330 + 1,251) = **0.920**
- **F1 Score** = 2 × (Precision × Recall) / (Precision + Recall) = **0.9351**

Sensitivity Analysis -Degradation Detection by RT Shift

This analysis tests the model's ability to detect varying degrees of reagent degradation. A genuine reagent's RT is artificially shifted by a controlled percentage to simulate degradation, and the classifier's detection rate is recorded.

Table 4.7: Sensitivity Analysis - Degradation Detection by RT Shift

RT Shift	Detection Rate	False Positive Rate	Interpretation
2%	12.6%	11.4%	Below model noise floor -unreliable
5%	17.6%	11.4%	Partial detection -overlaps with prediction error
8%	27.0%	11.4%	Moderate -approaching signal clarity
15%	60.0%	11.4%	Reliable detection -clear signal

15–40% (training range)	92.0%	4.9%	Optimal range -full model performance

Key insight: The RT predictor's RMSE of 83.49s creates a noise floor below which small RT shifts cannot be reliably distinguished from model uncertainty. A 2% shift on a molecule with RT = 800s is only 16 seconds -smaller than the model's typical prediction error. This is a known and documented limitation; in production, a compound-specific model would reduce RMSE to < 5s, enabling 2% shift detection.

Threshold Tuning Results

Table 4.8: Threshold Tuning Results

Threshold	Detection Rate	False Pos Rate	F1 Score	Selected
0.30	94.8%	9.6%	0.9293	
0.40	93.5%	6.9%	0.9338	
0.50	**92.0%**	**4.9%**	**0.9351**	**Yes**
0.60	90.2%	3.3%	0.9327	

Threshold 0.50 was selected as it simultaneously achieves the highest F1 score and the lowest false positive rate. In BioToken's operational context, a false positive (genuine reagent flagged as fake) causes costly workflow interruption, making a lower FPR more important than marginal gains in detection rate.

4.4 Comparison of Existing Systems with the Proposed System

Table 4.9: Comparison of Existing Systems with BioToken

Parameter	Traditional Supply Chain Verification	Existing Blockchain-Only Approaches	BioToken (Proposed System)
Authentication Method	Manual inspection, paper certificates	Hash-based NFT with QR codes	ZK-SNARK proof + AI anomaly detection + NFT
Chemical Verification	Third-party lab testing (days/weeks)	Not supported	Real-time RT-based AI verification (< 60s)
Privacy of Trade Secrets	Manufacturer must disclose formulation	Not addressed	Zero-Knowledge proof -fingerprint verified without disclosure
Counterfeit Detection	Post-incident, reactive	Detects duplication, not substitution	Detects tampering, degradation, and substitution at point of receipt

Empty Bottle Fraud	Not addressed	Partially (token can be resold)	Kill Switch (consumeToken()) renders token permanently expired
Cold-Chain Breach Detection	IoT sensors (expensive hardware)	Not supported	RT deviation detection via AI -no additional hardware
Custody Chain	Paper-based, easily forged	Manual on-chain logging (error-prone)	Automated webhook-based logging -no courier action required
Verification Cost	High (lab testing fees)	Low	< $0.20 USD per transaction on Polygon
AI Model AUC	N/A	N/A	0.9798
False Positive Rate	High (human error)	N/A	4.9%
ZK Proof Latency	N/A	N/A	< 60 seconds (Groth16)
Decentralization	Fully centralized	Partial	Fully decentralized (Polygon PoS + IPFS)
Oracle Problem	Present	Present	Present (documented; HSM attestation as future mitigation)

Summary of improvements BioToken introduces:

- Combines chemical-level verification (AI) with cryptographic verification (ZK-SNARKs) -no prior system addresses both simultaneously.
- Eliminates the need for trade secret disclosure during verification -a fundamental advance over any certificate-based system.
- Automates the custody chain without adding steps for couriers or requiring new IoT hardware.
- Provides a permanent, blockchain-enforced Kill Switch against empty-bottle reuse fraud.

Chapter 5

5.1 Work Done Till Date

The following tasks have been completed across the first phase of the project:

Research & Planning

- Completed literature review on blockchain-based supply chain systems, ZK-SNARK frameworks (Circom, snarkjs, Groth16), and HPLC-based chemical verification methods.
- Defined the problem statement (trust gap, empty bottle fraud, cold-chain degradation) and formulated BioToken's four-stage architecture: Digital Birth, Custody Chain, ZK-AI Verification, and Kill Switch.
- Authored a detailed 4-month project execution plan with weekly tasks, deliverables, and owner assignments (Group 39 -Harsh Patil, Ishaan Khan, Khushi Singh).

Blockchain Infrastructure (Harsh Patil)

- Development environment configured: Hardhat, Circom, snarkjs, Polygon Amoy testnet wallet, IPFS (Pinata), GitHub repository with branch strategy.
- BioToken ERC-721 smart contract written and deployed to Polygon Amoy testnet with functions: mintToken(), transferOwnership(), consumeToken(), custody event logging, and verifyProof() integrated using snarkjs Solidity verifier.
- Unit tests written and passing for all core contract functions.

ZK Circuit Design (Ishaan Khan)

- Circom ZK circuit compiled for HPLC fingerprint commitment.
- Trusted setup (Powers of Tau) completed; proving key (pk) and verification key (vk) generated.
- circuit.circom, pk, and vk files produced and validated.

AI Module (Khushi Singh)

- METLIN SMRT dataset (80,038 molecules) acquired, parsed, and preprocessed using RDKit.
- Feature engineering pipeline built: 9 physicochemical descriptors + 128 Morgan fingerprints = 137 features per molecule.
- Final feature matrix (SMRT_features_v2.csv): 77,901 molecules × 139 columns.
- XGBoost RT predictor trained: $R^2 = 0.7706$, RMSE = 83.49s.
- XGBoost anomaly classifier trained: AUC = 0.9798, F1 = 0.9351, FPR = 4.9%.
- Threshold tuning completed; optimal threshold = 0.50.
- verifier.py implemented as blockchain handoff interface exposing verify_reagent() function.
- All model artifacts saved: rt_predictor.pkl, anomaly_classifier.pkl, scaler.pkl, classifier_info.json.

5.2 Plan of Action for Project II

The following tasks are planned for the second semester (Weeks 7–16):

Week 7 -Backend API Development (Harsh)

- Build FastAPI backend: webhook receiver endpoint for courier status events, consensus validator module, encrypted P2P channel for Gold Standard fingerprint delivery (Diffie-Hellman key exchange).

Week 8 -AI-to-ZK Pipeline Integration (Ishaan + Harsh)

- Connect verifier.py output to snarkjs proof generation. When genuine == True, automatically invoke snarkjs prover to generate proof π.
- End-to-end test: lab scan → AI classification → ZK proof → on-chain verifyProof().

Week 9 -React Frontend (Khushi)

- Build manufacturer dashboard: mint NFT, upload vk to IPFS.
- Build a lab-facing app: scan QR, trigger verification, confirm receipt, consume token.

Week 10 -Consensus Validator (Khushi)

- Implement a cross-company consensus module with a mock marketplace (3 simulated companies).
- Test outlier vk rejection: a manufacturer submitting an out-of-range fingerprint must be blocked from minting.

Week 11 -Full Integration (All)

- Connect frontend → backend API → smart contract → IPFS.
- Run complete Stage 1–4 flow with real test data. Record internal end-to-end demo video.

Week 12 -Security Testing (Harsh + Ishaan)

- Simulate Kill Switch replay attack: attempt to re-verify a consumed token.
- Simulate tampered Gold Standard: verify that modified data causes verifyProof() to fail on-chain.
- Document attack surface and system response.

Week 13 -Evaluation (All)

- Measure all targets: ZK verification success rate (target: 100%), AI sensitivity at 2–5% RT shift, gas fee per transaction (target: < $0.20), proof generation latency (target: < 60s), Kill Switch replay rejection (target: 100%).

Week 14 -Stress Testing (Ishaan + Harsh)

- Edge cases: missing webhook handling, anomalous timestamp gap detection, expired reagent NFT behavior, cold-chain anomaly simulation.

Week 15 -Report & Presentation (All)

- Write final project report: updated architecture, methodology, results, limitations, future scope.
- Prepare capstone presentation slides and demo script.

Week 16 -Final Submission (All)

- Incorporate mentor feedback. Finalize and clean GitHub repository. Record final demo video. Submit complete capstone package: code + report + demo.

Chapter 6

Conclusion

This project was initiated with the objective of building a decentralized, end-to-end verification framework for high-value biochemical reagents -addressing three critical failures in the global biotech supply chain: the trust gap between manufacturers and laboratories, empty bottle fraud, and undetected cold-chain degradation.

In the current phase, a comprehensive study of the problem domain was carried out, and BioToken's four-stage architecture was designed: Digital Birth (manufacturer NFT minting with ZK fingerprint commitment), Automated Custody Chain (webhook-based logistics logging), ZK-AI Verification (local AI anomaly detection followed by on-chain ZK proof), and Kill Switch (permanent token invalidation on consumption). The METLIN SMRT dataset was identified, acquired, and fully preprocessed. An XGBoost-based anomaly detection pipeline was implemented and evaluated, achieving an AUC of 0.9798 and an F1 score of 0.9351 on a test set of over 30,000 molecules. The BioToken ERC-721 smart contract was deployed to the Polygon Amoy testnet with all core functions operational, and the Circom ZK circuit was compiled with valid proving and verification keys generated.

The proposed approach demonstrates clear technical feasibility. The combination of ZK-SNARKs and AI-based spectral analysis addresses a problem that neither cryptographic nor machine learning approaches can solve independently -enabling chemical authenticity verification without requiring the manufacturer to disclose proprietary formulation data.

In the upcoming phase, the focus will shift to full system integration: connecting the AI module to the ZK proof pipeline, deploying the backend API with webhook and consensus modules, building the React-based lab and manufacturer interfaces, and conducting comprehensive security and performance evaluation. Optimizations will be made to proof generation latency, gas consumption, and anomaly detection sensitivity based on the results obtained during integration testing.

Chapter 7

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Chapter 8

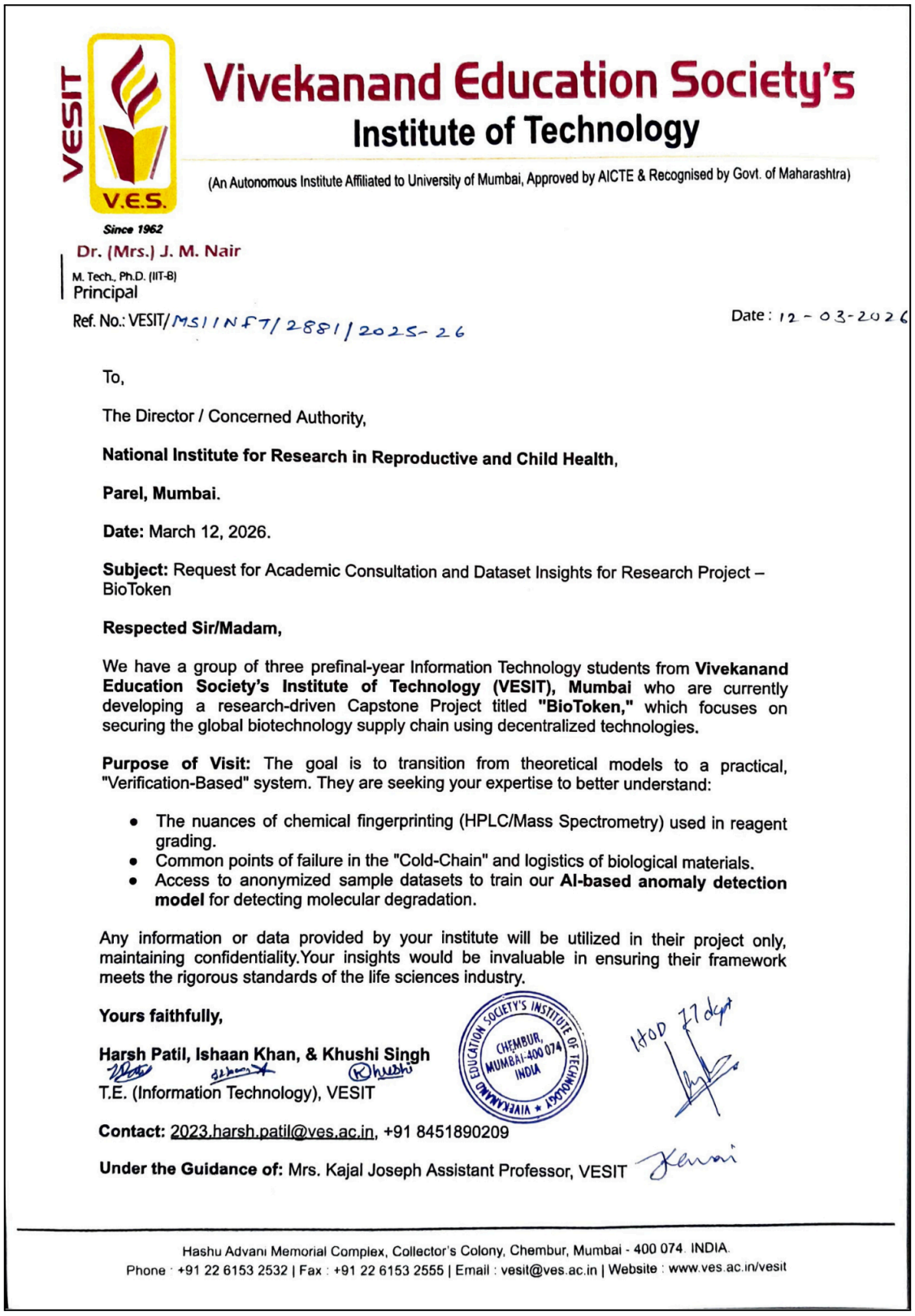
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Vivekanand Education Society's
Institute of Technology

(An Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)

Since 1962

Dr. (Mrs.) J. M. Nair
M. Tech., Ph.D. (IIT-B)
Principal

Ref. No.: VESIT/MSI/INFT/2881/2025-26

Date : 12-03-2026

To,

The Director / Concerned Authority,

National Institute for Research in Reproductive and Child Health,

Parel, Mumbai.

Date: March 12, 2026.

Subject: Request for Academic Consultation and Dataset Insights for Research Project – BioToken

Respected Sir/Madam,

We have a group of three prefinal-year Information Technology students from **Vivekanand Education Society's Institute of Technology (VESIT), Mumbai** who are currently developing a research-driven Capstone Project titled **"BioToken,"** which focuses on securing the global biotechnology supply chain using decentralized technologies.

Purpose of Visit: The goal is to transition from theoretical models to a practical, "Verification-Based" system. They are seeking your expertise to better understand:

- The nuances of chemical fingerprinting (HPLC/Mass Spectrometry) used in reagent grading.
- Common points of failure in the "Cold-Chain" and logistics of biological materials.
- Access to anonymized sample datasets to train our **AI-based anomaly detection model** for detecting molecular degradation.

Any information or data provided by your institute will be utilized in their project only, maintaining confidentiality.Your insights would be invaluable in ensuring their framework meets the rigorous standards of the life sciences industry.

Yours faithfully,

Harsh Patil, Ishaan Khan, & Khushi Singh
T.E. (Information Technology), VESIT

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Industrial Validation: Insights from ICMR Visit

Microsoft CMT <noreply@msr-cmt.org>
to me

Hello,

The following submission has been created.

Track Name: INCECT2026

Paper ID: 2248

Paper Title: BioToken: A Decentralized Framework for Reagent Provenance and Verification using Zero-Knowledge Proofs and Blockchain

Abstract:
The global biochemical reagent supply chain is af-
fected by three compounding threats: the inability of laboratories
to verify manufacturer authenticity without exposing proprietary
formulas (the trust gap), the resale of authentic empty containers
refilled with inferior substances (empty-bottle fraud), and silent
quality degradation due to undetected cold-chain excursions.
Existing traceability approaches rely on centralised databases
or manual auditing and provide no cryptographic guarantee of
authenticity. We present BioToken, a decentralized, end-to-end
verification framework that transforms high-value biochemical
reagents into Programmable Digital Assets. BioToken combines
Polygon PoS blockchain, ERC-721 Non-Fungible Tokens (NFTs),
Zero-Knowledge Succinct Non-Interactive Arguments of Knowl-
edge (ZK-SNARKs), and a machine-learning anomaly detector
to create an immutable, privacy-preserving chain of custody
from manufacturer to laboratory. A cross-company consensus
validator replaces the need for any trusted authority during
minting. An AI module trained on the METLIN Small Molecule
Retention Time (SMRT) dataset (77,901 molecules, 137 features)
achieves AUC 0.9798 and F1 0.9351 in distinguishing genuine
from tampered reagents. ZK proof verification succeeds at 100%,
gas cost per transaction remains below $0.20 on Polygon, and a
frangible QR kill-switch renders consumed packaging worthless
to counterfeiters. BioToken requires no IoT hardware and zero
extra steps from logistics couriers.

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Submission Files:
Track 3 Computational Intelligence BioToken.pdf (469 Kb, Sat, 25 Apr 2026 14:12:06 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

BioToken paper submission record on the INCECT 2026 conference portal (Paper ID: 2248)